

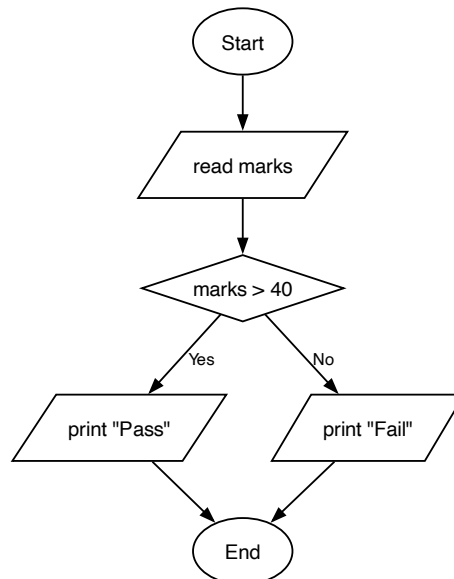
Name: _____

Date: _____

In every problem below, the flowchart has **exactly one** wrong box. The words in italics say what the program is *supposed* to do; the table under the flowchart shows what it *should* print and what it *actually* prints. Work backwards from that difference: do not redraw the whole flowchart — find the single box that is wrong, write what it should say instead, and explain why that one box produces the wrong output.

Problem 1.

A student passes if they score 40 marks or more. The program reads a student's marks and prints "Pass" or "Fail".



input	should print	actually prints
marks = 72	Pass	Pass
marks = 40	Pass	Fail
marks = 12	Fail	Fail

(i) Which box is wrong? Copy it here: _____

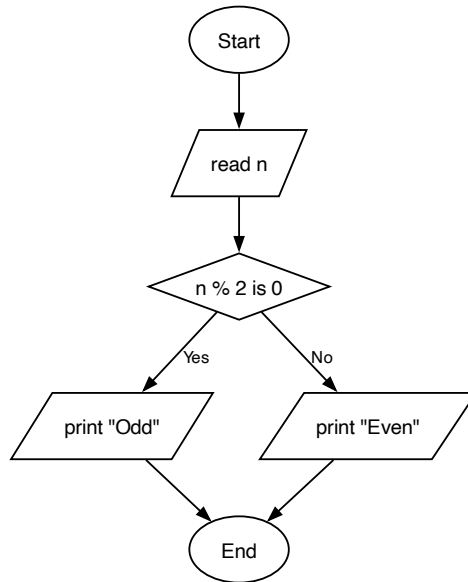
(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 2.

The program reads a number and prints “Even” if the number is even, and “Odd” if it is odd.

Reminder: $n \% 2$ is the remainder when n is divided by 2, so $n \% 2$ is 0 exactly when n is even.



input	should print	actually prints
n = 6	Even	Odd
n = 7	Odd	Even

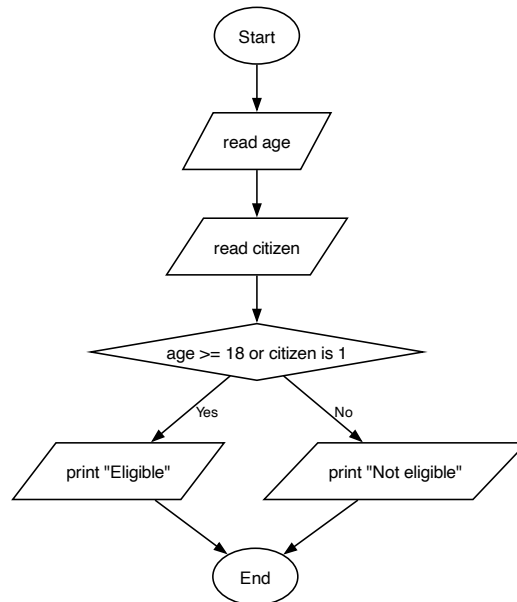
(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 3.

A person may vote only if they are 18 or older AND they are a citizen. The program reads the person's age and citizen (1 means yes, 0 means no) and prints "Eligible" or "Not eligible".



input	should print	actually prints
age = 20, citizen = 1	Eligible	Eligible
age = 15, citizen = 1	Not eligible	Eligible
age = 30, citizen = 0	Not eligible	Eligible

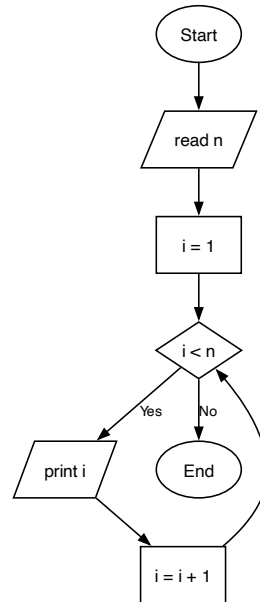
(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 4.

The program reads a number n and prints every number from 1 up to n — so for $n = 5$ it prints 1, 2, 3, 4, 5.



input	should print	actually prints
$n = 5$	1, 2, 3, 4, 5	1, 2, 3, 4
$n = 1$	1	nothing

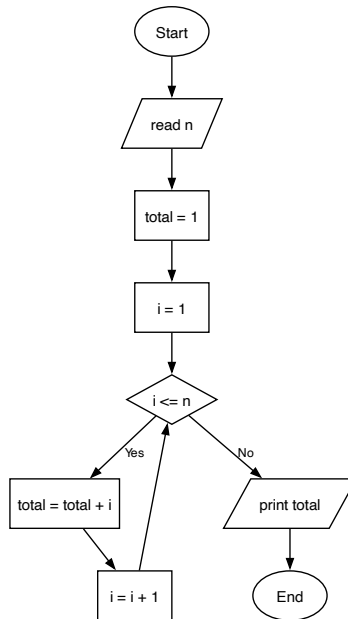
(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 5.

The program reads a number n and prints the total of all the numbers from 1 to n — so for $n = 5$ it prints $1+2+3+4+5 = 15$.



input	should print	actually prints
$n = 5$	15	16
$n = 3$	6	7

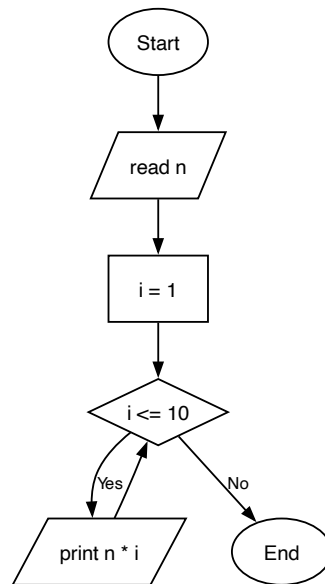
(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 6.

The program reads a number n and prints its multiplication table from $n \times 1$ up to $n \times 10$ — so for $n = 3$ it prints 3, 6, 9, ..., 30.



input	should print	actually prints
$n = 3$	3, 6, 9, 12, 15, 18, 21, 24, 27, 30	3, 3, 3, 3, 3, ... <i>(never stops)</i>

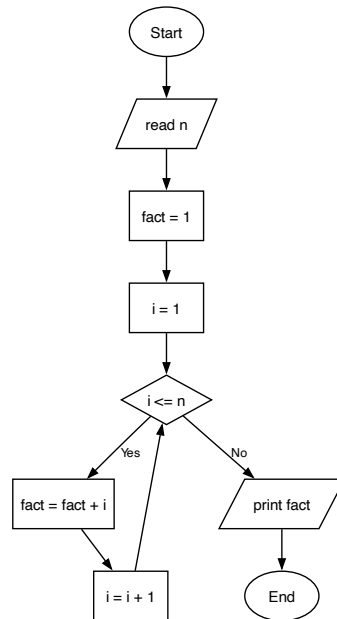
(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 7.

The program reads a number n and prints its factorial — the product $1 \times 2 \times 3 \times \dots \times n$. So for $n = 4$ it prints $1 \times 2 \times 3 \times 4 = 24$.



input	should print	actually prints
$n = 4$	24	11
$n = 5$	120	16

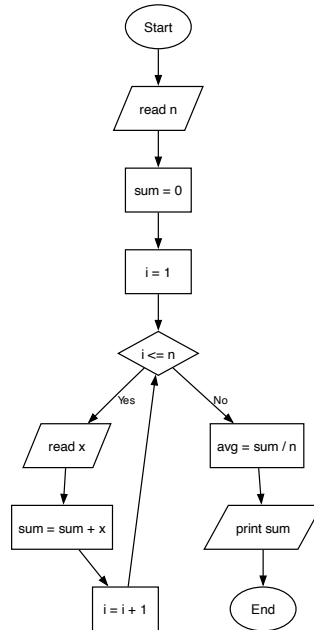
(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 8.

The program reads how many numbers there are (n), then reads that many numbers one by one and prints their average — the total divided by how many there were.



input	should print	actually prints
n = 4, x = 10, 20, 30, 40	25	100
n = 2, x = 7, 9	8	16

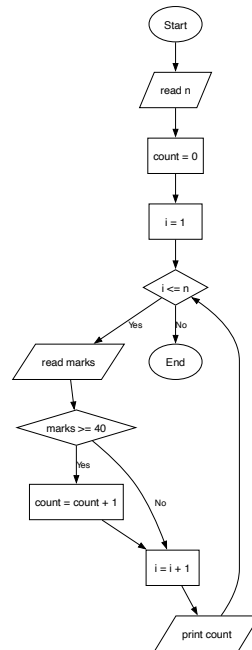
(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 9.

The program reads how many students there are (n), then reads each student's marks and finally prints how many of them passed (40 marks or more). For the marks 55, 32, 80, 40 it should print just one number: 3.



input	should print	actually prints
n = 4, marks = 55, 32, 80, 40	3	1, 1, 2, 3

(i) Which box is wrong? Copy it here: _____

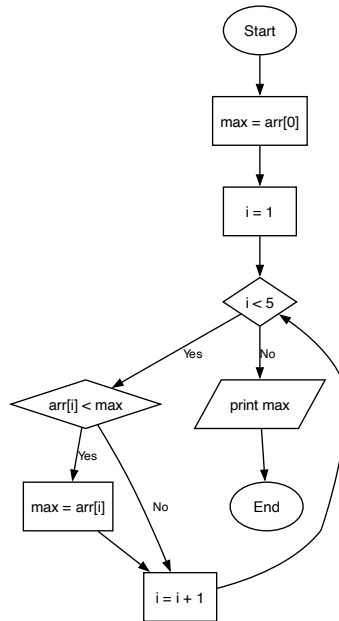
(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?

Problem 10.

The program goes through a list of 5 numbers and prints the largest of them. It starts by assuming the first item is the biggest, then checks every later item and remembers any item that beats the best one so far.

Reminder: the items of a list have positions counted from 0, so the list [3, 9, 2, 12, 7] has 5 items in positions 0 to 4. $\text{arr}[i]$ means the item at position i , so $\text{arr}[0]$ is 3.



input	should print	actually prints
arr = 3, 9, 2, 12, 7	12	2
arr = 8, 1, 6, 4, 5	8	1

(i) Which box is wrong? Copy it here: _____

(ii) What should that box say instead? _____

(iii) Why does that mistake produce the wrong output?
